

Leading asymptotic of the standard integral for general continuous data

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Abstract

The one-dimensional leading term of `cor:standardintegralexp` in its natural generality: for any continuous ξ, η on the chart with $|\xi(u) - \xi_0| \leq Cu$ and $|\eta(u) - \eta_0| \leq C'u$ (any C^1 or analytic function at the origin), and $\eta_0 \neq 0$,

$$\int_0^b u^h \eta(u) e^{-\beta n u^{2k} + \beta \sqrt{n} u^k \xi(u)} du \sim \eta_0 \frac{1}{2k} S_{(h+1)/(2k)}(\xi_0) n^{-(h+1)/(2k)},$$

with difference $O(n^{-(h+2)/(2k)})$. No series expansion of ξ is required; this supersedes the polynomial special cases (units 27, 33) for the leading order. Zero `sorry/axiom`.

1 Setting

Unit 33 obtained the leading term for polynomial data by expanding ξ and η and bounding every non-leading monomial. Looking back at that proof, only one property of the expansion was used at leading order: $|\xi(u) - \xi_0| \leq Cu$ (and likewise for η). Making that the hypothesis gives a cleaner and far more general theorem, with the first-order remainder $e^x - 1$, $x = \beta \sqrt{n} u^k (\xi(u) - \xi_0)$, controlled by $|e^x - 1| \leq |x|e^{|x|}$ — the $N = 1$ case of the global exponential remainder bound of unit 30.

2 The things formalised

```
theorem abs_exp_sub_one_le_abs_mul_exp (x : ℝ) : |Real.exp x - 1| ≤ |x| * Real.exp |x|
```

```
theorem standardIntegral1D_leading_isBigO_of_lipschitz (b C C' : ℝ) (h : ℝ → ℝ) (k : ℕ) (hb : 0 < b) (hk : 0 < k) (hC : 0 < C) (hC' : Continuous C') (hC : Continuous C) (h : u ↦ Ioc 0 b, |u - | C * u) (h : u ↦ Ioc 0 b, |u - | C' * u) : (fun n => Z_chart(, ) n - * ((1/(2k)) * S_{(h+1)/(2k)}( )) * n^{-(h+1)/(2k)}) =0[atTop] fun n => n ^ (-(h:ℕ)+2)/(2k))
```

```
theorem standardIntegral1D_leading_isEquivalent_of_lipschitz ... (h : ℝ → ℝ) ... : (fun n => Z_chart(, ) n) ~[atTop] fun n => * ((1/(2k)) * S_{(h+1)/(2k)}( )) * n^{-(h+1)/(2k)}
```

3 Proof strategy

Pointwise, $u^h \eta e^{\text{pop}} e^x = \eta_0 u^h e^{\text{pop}} + \eta_0 u^h e^{\text{pop}} (e^x - 1) + u^h (\eta - \eta_0) e^{\text{pop}} e^x$. The first piece is η_0 times the half-line value minus its exponentially small tail (units 22, 25). For the second, $|e^x - 1| \leq |x|e^{|x|}$

with $|x| \leq \beta\sqrt{n}u^kCu$ and $e^{\text{pop}}e^{|x|} \leq e^{\text{pop}}$ at the shifted argument $\xi_0 + Cb$ gives the majorant $|\eta_0|\beta C u^{h+1}(\sqrt{n}u^k)e^{\text{pop}'}$, a $p = 1$ insertion at $h + 1$: order $n^{-(h+2)/(2k)}$. For the third, $|\eta - \eta_0| \leq C'u$ and $e^x \leq e^{|x|}$ give $C'u^{h+1}e^{\text{pop}'}$, the kernel identity at $h + 1$: again $n^{-(h+2)/(2k)}$. `IsBig0.of_bound` with the sum of the three constants, then the power little- o and `IsLittle0.const_mul_right` for the equivalence.

4 Roadblocks and resolutions

None: first-try compile, using the idioms accumulated over units 28–33 (opaque local definitions, explicit `mul_le_mul` chains instead of `gcongr` on `rpow`, a universally quantified `key` identity for the final rearrangement, the insertion identity as the sole evaluator). A small point: $0 \leq C'$ is not assumed but derived from the hypothesis at $u = b$.

5 What was Mathlib, what was new

Mathlib: `Finset.sum_range_one`, `abs_integral_le_integral_abs`, `setIntegral_mono_on/mono_set`, `isLittle0_exp_neg_mul_rpow_atTop`. New: the leading asymptotic under a Lipschitz-at-the-origin hypothesis, and the global $|e^x - 1| \leq |x|e^{|x|}$.

6 Lessons

After proving a result for polynomials, ask which property of the polynomial the proof actually used; here it was a single linear bound at the origin, and stating that as the hypothesis both generalised the theorem and shortened the proof.

7 Outcome

The one-dimensional leading-order theory of grammar §4.2 is complete in natural generality; the higher-order expansions are available for monomial and (to second order) polynomial perturbations. The remaining gaps are the higher-order terms for general analytic ξ (the tree combinatorics) and the multivariate theory.